

Question 9

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9. Do as follows using in-built “USArrests” dataset with R script to knit PDF output.

a. Create a “criminality scale” of four variables of this dataset using the Principal Component Analysis

```
# Load required libraries
```

```
library(psych)
```

```
library(ggplot2)
```

```
##
```

```
## Attaching package: 'ggplot2'
```

```
## The following objects are masked from 'package:psych':
```

```
##
```

```
##      %+%, alpha
```

```
# Perform PCA on USArrests dataset
```

```
pca_result <- prcomp(USArrests, scale = TRUE)
```

```
# Display summary of PCA results
```

```
summary(pca_result)
```

```
## Importance of components:
```

```
##              PC1      PC2      PC3      PC4
```

```
## Standard deviation    1.5749 0.9949 0.59713 0.41645
```

```
## Proportion of Variance 0.6201 0.2474 0.08914 0.04336
```

```
## Cumulative Proportion 0.6201 0.8675 0.95664 1.00000
```

b. Compute the eigenvalues and interpret the PCA result carefully using Kaiser’s criteria

```
# Extract eigenvalues from PCA results
```

```
eigenvalues <- pca_result$sdev^2
```

```
# Create a data frame for eigenvalues
```

```
eigen_df <- data.frame(
```

```
  Component = 1:length(eigenvalues),
```

```
  Eigenvalue = eigenvalues
```

```
)
```

```
# Display eigenvalues
```

```
print("Eigenvalues:")
```

```
## [1] "Eigenvalues:"
```

```
print(eigenvalues)
```

```
## [1] 2.4802416 0.9897652 0.3565632 0.1734301
```

```
# Kaiser's criterion (eigenvalues > 1)
```

```
print("Components meeting Kaiser's criterion (eigenvalue > 1):")
```

```
## [1] "Components meeting Kaiser's criterion (eigenvalue > 1):"
```

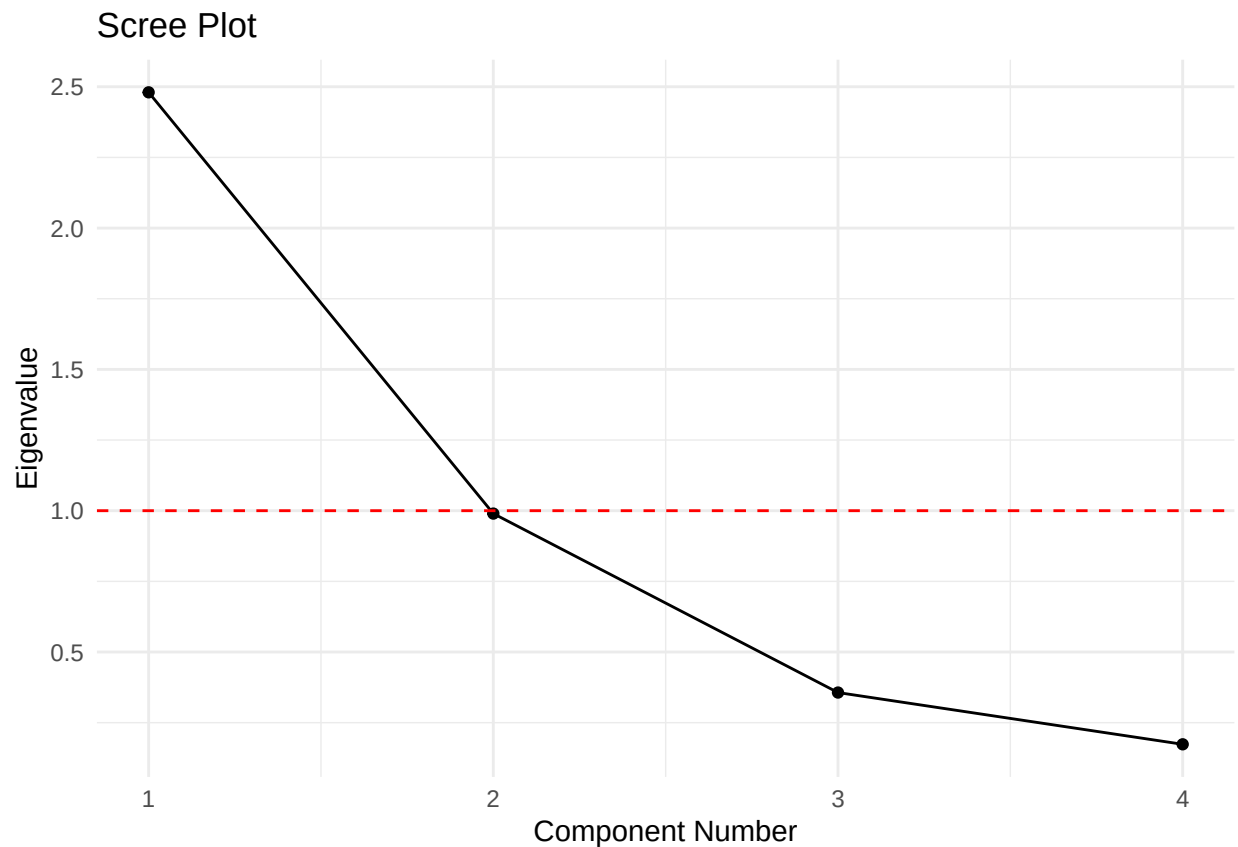
```
print(sum(eigenvalues > 1))
```

```
## [1] 1
```

c. Show the Scree plot and decide the number of components to retain with careful interpretation

```
# Create Scree plot
```

```
ggplot(eigen_df, aes(x = Component, y = Eigenvalue)) +  
  geom_line() +  
  geom_point() +  
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +  
  labs(title = "Scree Plot",  
       x = "Component Number",  
       y = "Eigenvalue") +  
  theme_minimal()
```



d. Revise the criminality scale using varimax rotation and interpret the results carefully

```
# Perform PCA with varimax rotation
pca_rotated <- principal(USArrests, nfactors = 2, rotate = "varimax")
```

```
# Display rotated component loadings
print("Rotated Component Loadings:")
```

```
## [1] "Rotated Component Loadings:"
```

```
print(pca_rotated$loadings, cutoff = 0.3)
```

```
##
## Loadings:
##          RC1      RC2
## Murder    0.939
## Assault   0.920
## UrbanPop          0.970
## Rape      0.727  0.481
##
##          RC1      RC2
## SS loadings  2.262  1.208
## Proportion Var 0.565  0.302
## Cumulative Var 0.565  0.868
```

```
# Display variance explained by rotated components
print("Variance Explained by Rotated Components:")
```

```
## [1] "Variance Explained by Rotated Components:"
```

```
print(pca_rotated$Vaccounted)
```

```
##
##          RC1      RC2
## SS loadings  2.2617016  1.2083051
## Proportion Var  0.5654254  0.3020763
## Cumulative Var  0.5654254  0.8675017
## Proportion Explained  0.6517859  0.3482141
## Cumulative Proportion 0.6517859  1.0000000
```